108 Finger Permutations - Level 11

Three-Position Relay: 1+2+1 Double-Shift Control

Our lesson plans go to 11.

Levels 8-10 covered every useful single-shift boundary. Level 11 closes the position branch with two relocations inside every code.

1. Keep three frames exact

P5: 1=5, 2=6, 3=7, 4=8.
P7: 1=7, 2=8, 3=9, 4=10.
P9: 1=9, 2=10, 3=11, 4=12.
Move the hand and thumb as one unit.
Do not stretch across position frames.

2. Relay every code 1+2+1

Note 1: first position frame.
Notes 2-3: middle position frame.
Note 4: third position frame.
UP: P5>P7>P9. DOWN: P9>P7>P5.
The four-digit finger code never changes.

3. Control both landing notes

Landing notes are 2 and 4.
Alternate picking never resets at a shift.
Remove pause, slide noise, and extra accent.
Pitch may rise, repeat, or fall while the hand frame still moves in the printed direction.

Worked example - code 1234

Each vertical bar is a whole-frame relocation. Repeat the one-string template sequentially on every string in the printed route.

SHIFT UP - E>A>D>G>B>e P5>P7>P9

```
F|1------|2-3----|4------|
S|5------|8-9----|12-----|
```

SHIFT DOWN - e>B>G>D>A>E P9>P7>P5

```
F|1------|2-3----|4------|
S|9------|8-9----|8------|
```

In the DOWN example, fret 8 appears in both P7 and P5. The printed frame and assigned finger remain authoritative.

LANDINGS Lightly accent notes 2 and 4 only while learning the two relocations.

PICKING `D-start  D | U-D | U  U-start  U | D-U | D`

Do not restart, reverse, or exaggerate the picking cycle at either shift boundary.

AFTER MASTERY

Keep the 1+2+1 grouping and vary only the distance: P5>P8>P11 or P5>P7>P10, then reverse.
These are distance variants, not another 108-exercise level.

Session protocol

1. Choose one 12-exercise set.
2. Begin 10-15 BPM below the slower of your two clean Level 10 modes.
3. Pass 1: use even eighths, pause at both bars, and accent notes 2 and 4.
4. Pass 2: remove both pauses, then remove both landing accents.
5. Run UP and DOWN with both pick starts; use the +2 / -4-6 BPM rule.
6. Advance after 2 clean days; then restore triplet and quintuplet pulses.

Level 11 - Three-Position Relay

Set 1 of 9 - Exercises 001-012

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

001 1212

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-1----|2------|
S|5------|8-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-1----|2------|
S|9------|8-7----|6------|
```

landings 2 and 4; no pick reset

002 1213

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-1----|3------|
S|5------|8-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-1----|3------|
S|9------|8-7----|7------|
```

landings 2 and 4; no pick reset

003 1214

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-1----|4------|
S|5------|8-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-1----|4------|
S|9------|8-7----|8------|
```

landings 2 and 4; no pick reset

004 1231

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-3----|1------|
S|5------|8-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-3----|1------|
S|9------|8-9----|5------|
```

landings 2 and 4; no pick reset

005 1232

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-3----|2------|
S|5------|8-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-3----|2------|
S|9------|8-9----|6------|
```

landings 2 and 4; no pick reset

006 1234

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-3----|4------|
S|5------|8-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-3----|4------|
S|9------|8-9----|8------|
```

landings 2 and 4; no pick reset

007 1241

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-4----|1------|
S|5------|8-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-4----|1------|
S|9------|8-10---|5------|
```

landings 2 and 4; no pick reset

008 1242

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-4----|2------|
S|5------|8-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-4----|2------|
S|9------|8-10---|6------|
```

landings 2 and 4; no pick reset

009 1243

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|2-4----|3------|
S|5------|8-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|2-4----|3------|
S|9------|8-10---|7------|
```

landings 2 and 4; no pick reset

010 1312

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-1----|2------|
S|5------|9-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-1----|2------|
S|9------|9-7----|6------|
```

landings 2 and 4; no pick reset

011 1313

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-1----|3------|
S|5------|9-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-1----|3------|
S|9------|9-7----|7------|
```

landings 2 and 4; no pick reset

012 1314

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-1----|4------|
S|5------|9-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-1----|4------|
S|9------|9-7----|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 2 of 9 - Exercises 013-024

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

013 1321

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-2----|1------|
S|5------|9-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-2----|1------|
S|9------|9-8----|5------|
```

landings 2 and 4; no pick reset

014 1323

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-2----|3------|
S|5------|9-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-2----|3------|
S|9------|9-8----|7------|
```

landings 2 and 4; no pick reset

015 1324

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-2----|4------|
S|5------|9-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-2----|4------|
S|9------|9-8----|8------|
```

landings 2 and 4; no pick reset

016 1341

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-4----|1------|
S|5------|9-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-4----|1------|
S|9------|9-10---|5------|
```

landings 2 and 4; no pick reset

017 1342

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-4----|2------|
S|5------|9-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-4----|2------|
S|9------|9-10---|6------|
```

landings 2 and 4; no pick reset

018 1343

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|3-4----|3------|
S|5------|9-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|3-4----|3------|
S|9------|9-10---|7------|
```

landings 2 and 4; no pick reset

019 1412

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-1----|2------|
S|5------|10-7---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-1----|2------|
S|9------|10-7---|6------|
```

landings 2 and 4; no pick reset

020 1413

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-1----|3------|
S|5------|10-7---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-1----|3------|
S|9------|10-7---|7------|
```

landings 2 and 4; no pick reset

021 1414

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-1----|4------|
S|5------|10-7---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-1----|4------|
S|9------|10-7---|8------|
```

landings 2 and 4; no pick reset

022 1421

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-2----|1------|
S|5------|10-8---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-2----|1------|
S|9------|10-8---|5------|
```

landings 2 and 4; no pick reset

023 1423

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-2----|3------|
S|5------|10-8---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-2----|3------|
S|9------|10-8---|7------|
```

landings 2 and 4; no pick reset

024 1424

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-2----|4------|
S|5------|10-8---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-2----|4------|
S|9------|10-8---|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 3 of 9 - Exercises 025-036

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

025 1431

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-3----|1------|
S|5------|10-9---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-3----|1------|
S|9------|10-9---|5------|
```

landings 2 and 4; no pick reset

026 1432

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-3----|2------|
S|5------|10-9---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-3----|2------|
S|9------|10-9---|6------|
```

landings 2 and 4; no pick reset

027 1434

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|1------|4-3----|4------|
S|5------|10-9---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|1------|4-3----|4------|
S|9------|10-9---|8------|
```

landings 2 and 4; no pick reset

028 2121

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-2----|1------|
S|6------|7-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-2----|1------|
S|10-----|7-8----|5------|
```

landings 2 and 4; no pick reset

029 2123

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-2----|3------|
S|6------|7-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-2----|3------|
S|10-----|7-8----|7------|
```

landings 2 and 4; no pick reset

030 2124

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-2----|4------|
S|6------|7-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-2----|4------|
S|10-----|7-8----|8------|
```

landings 2 and 4; no pick reset

031 2131

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-3----|1------|
S|6------|7-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-3----|1------|
S|10-----|7-9----|5------|
```

landings 2 and 4; no pick reset

032 2132

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-3----|2------|
S|6------|7-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-3----|2------|
S|10-----|7-9----|6------|
```

landings 2 and 4; no pick reset

033 2134

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-3----|4------|
S|6------|7-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-3----|4------|
S|10-----|7-9----|8------|
```

landings 2 and 4; no pick reset

034 2141

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-4----|1------|
S|6------|7-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-4----|1------|
S|10-----|7-10---|5------|
```

landings 2 and 4; no pick reset

035 2142

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-4----|2------|
S|6------|7-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-4----|2------|
S|10-----|7-10---|6------|
```

landings 2 and 4; no pick reset

036 2143

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|1-4----|3------|
S|6------|7-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|1-4----|3------|
S|10-----|7-10---|7------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 4 of 9 - Exercises 037-048

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

037 2312

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-1----|2------|
S|6------|9-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-1----|2------|
S|10-----|9-7----|6------|
```

landings 2 and 4; no pick reset

038 2313

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-1----|3------|
S|6------|9-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-1----|3------|
S|10-----|9-7----|7------|
```

landings 2 and 4; no pick reset

039 2314

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-1----|4------|
S|6------|9-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-1----|4------|
S|10-----|9-7----|8------|
```

landings 2 and 4; no pick reset

040 2321

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-2----|1------|
S|6------|9-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-2----|1------|
S|10-----|9-8----|5------|
```

landings 2 and 4; no pick reset

041 2323

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-2----|3------|
S|6------|9-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-2----|3------|
S|10-----|9-8----|7------|
```

landings 2 and 4; no pick reset

042 2324

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-2----|4------|
S|6------|9-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-2----|4------|
S|10-----|9-8----|8------|
```

landings 2 and 4; no pick reset

043 2341

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-4----|1------|
S|6------|9-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-4----|1------|
S|10-----|9-10---|5------|
```

landings 2 and 4; no pick reset

044 2342

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-4----|2------|
S|6------|9-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-4----|2------|
S|10-----|9-10---|6------|
```

landings 2 and 4; no pick reset

045 2343

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|3-4----|3------|
S|6------|9-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|3-4----|3------|
S|10-----|9-10---|7------|
```

landings 2 and 4; no pick reset

046 2412

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-1----|2------|
S|6------|10-7---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-1----|2------|
S|10-----|10-7---|6------|
```

landings 2 and 4; no pick reset

047 2413

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-1----|3------|
S|6------|10-7---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-1----|3------|
S|10-----|10-7---|7------|
```

landings 2 and 4; no pick reset

048 2414

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-1----|4------|
S|6------|10-7---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-1----|4------|
S|10-----|10-7---|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 5 of 9 - Exercises 049-060

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

049 2421

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-2----|1------|
S|6------|10-8---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-2----|1------|
S|10-----|10-8---|5------|
```

landings 2 and 4; no pick reset

050 2423

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-2----|3------|
S|6------|10-8---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-2----|3------|
S|10-----|10-8---|7------|
```

landings 2 and 4; no pick reset

051 2424

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-2----|4------|
S|6------|10-8---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-2----|4------|
S|10-----|10-8---|8------|
```

landings 2 and 4; no pick reset

052 2431

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-3----|1------|
S|6------|10-9---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-3----|1------|
S|10-----|10-9---|5------|
```

landings 2 and 4; no pick reset

053 2432

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-3----|2------|
S|6------|10-9---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-3----|2------|
S|10-----|10-9---|6------|
```

landings 2 and 4; no pick reset

054 2434

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|2------|4-3----|4------|
S|6------|10-9---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|2------|4-3----|4------|
S|10-----|10-9---|8------|
```

landings 2 and 4; no pick reset

055 3121

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-2----|1------|
S|7------|7-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-2----|1------|
S|11-----|7-8----|5------|
```

landings 2 and 4; no pick reset

056 3123

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-2----|3------|
S|7------|7-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-2----|3------|
S|11-----|7-8----|7------|
```

landings 2 and 4; no pick reset

057 3124

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-2----|4------|
S|7------|7-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-2----|4------|
S|11-----|7-8----|8------|
```

landings 2 and 4; no pick reset

058 3131

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-3----|1------|
S|7------|7-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-3----|1------|
S|11-----|7-9----|5------|
```

landings 2 and 4; no pick reset

059 3132

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-3----|2------|
S|7------|7-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-3----|2------|
S|11-----|7-9----|6------|
```

landings 2 and 4; no pick reset

060 3134

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-3----|4------|
S|7------|7-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-3----|4------|
S|11-----|7-9----|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 6 of 9 - Exercises 061-072

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

061 3141

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-4----|1------|
S|7------|7-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-4----|1------|
S|11-----|7-10---|5------|
```

landings 2 and 4; no pick reset

062 3142

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-4----|2------|
S|7------|7-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-4----|2------|
S|11-----|7-10---|6------|
```

landings 2 and 4; no pick reset

063 3143

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|1-4----|3------|
S|7------|7-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|1-4----|3------|
S|11-----|7-10---|7------|
```

landings 2 and 4; no pick reset

064 3212

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-1----|2------|
S|7------|8-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-1----|2------|
S|11-----|8-7----|6------|
```

landings 2 and 4; no pick reset

065 3213

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-1----|3------|
S|7------|8-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-1----|3------|
S|11-----|8-7----|7------|
```

landings 2 and 4; no pick reset

066 3214

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-1----|4------|
S|7------|8-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-1----|4------|
S|11-----|8-7----|8------|
```

landings 2 and 4; no pick reset

067 3231

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-3----|1------|
S|7------|8-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-3----|1------|
S|11-----|8-9----|5------|
```

landings 2 and 4; no pick reset

068 3232

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-3----|2------|
S|7------|8-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-3----|2------|
S|11-----|8-9----|6------|
```

landings 2 and 4; no pick reset

069 3234

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-3----|4------|
S|7------|8-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-3----|4------|
S|11-----|8-9----|8------|
```

landings 2 and 4; no pick reset

070 3241

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-4----|1------|
S|7------|8-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-4----|1------|
S|11-----|8-10---|5------|
```

landings 2 and 4; no pick reset

071 3242

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-4----|2------|
S|7------|8-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-4----|2------|
S|11-----|8-10---|6------|
```

landings 2 and 4; no pick reset

072 3243

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|2-4----|3------|
S|7------|8-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|2-4----|3------|
S|11-----|8-10---|7------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 7 of 9 - Exercises 073-084

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

073 3412

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-1----|2------|
S|7------|10-7---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-1----|2------|
S|11-----|10-7---|6------|
```

landings 2 and 4; no pick reset

074 3413

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-1----|3------|
S|7------|10-7---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-1----|3------|
S|11-----|10-7---|7------|
```

landings 2 and 4; no pick reset

075 3414

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-1----|4------|
S|7------|10-7---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-1----|4------|
S|11-----|10-7---|8------|
```

landings 2 and 4; no pick reset

076 3421

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-2----|1------|
S|7------|10-8---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-2----|1------|
S|11-----|10-8---|5------|
```

landings 2 and 4; no pick reset

077 3423

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-2----|3------|
S|7------|10-8---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-2----|3------|
S|11-----|10-8---|7------|
```

landings 2 and 4; no pick reset

078 3424

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-2----|4------|
S|7------|10-8---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-2----|4------|
S|11-----|10-8---|8------|
```

landings 2 and 4; no pick reset

079 3431

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-3----|1------|
S|7------|10-9---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-3----|1------|
S|11-----|10-9---|5------|
```

landings 2 and 4; no pick reset

080 3432

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-3----|2------|
S|7------|10-9---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-3----|2------|
S|11-----|10-9---|6------|
```

landings 2 and 4; no pick reset

081 3434

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|3------|4-3----|4------|
S|7------|10-9---|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|3------|4-3----|4------|
S|11-----|10-9---|8------|
```

landings 2 and 4; no pick reset

082 4121

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-2----|1------|
S|8------|7-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-2----|1------|
S|12-----|7-8----|5------|
```

landings 2 and 4; no pick reset

083 4123

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-2----|3------|
S|8------|7-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-2----|3------|
S|12-----|7-8----|7------|
```

landings 2 and 4; no pick reset

084 4124

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-2----|4------|
S|8------|7-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-2----|4------|
S|12-----|7-8----|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 8 of 9 - Exercises 085-096

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

085 4131

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-3----|1------|
S|8------|7-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-3----|1------|
S|12-----|7-9----|5------|
```

landings 2 and 4; no pick reset

086 4132

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-3----|2------|
S|8------|7-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-3----|2------|
S|12-----|7-9----|6------|
```

landings 2 and 4; no pick reset

087 4134

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-3----|4------|
S|8------|7-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-3----|4------|
S|12-----|7-9----|8------|
```

landings 2 and 4; no pick reset

088 4141

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-4----|1------|
S|8------|7-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-4----|1------|
S|12-----|7-10---|5------|
```

landings 2 and 4; no pick reset

089 4142

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-4----|2------|
S|8------|7-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-4----|2------|
S|12-----|7-10---|6------|
```

landings 2 and 4; no pick reset

090 4143

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|1-4----|3------|
S|8------|7-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|1-4----|3------|
S|12-----|7-10---|7------|
```

landings 2 and 4; no pick reset

091 4212

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-1----|2------|
S|8------|8-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-1----|2------|
S|12-----|8-7----|6------|
```

landings 2 and 4; no pick reset

092 4213

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-1----|3------|
S|8------|8-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-1----|3------|
S|12-----|8-7----|7------|
```

landings 2 and 4; no pick reset

093 4214

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-1----|4------|
S|8------|8-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-1----|4------|
S|12-----|8-7----|8------|
```

landings 2 and 4; no pick reset

094 4231

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-3----|1------|
S|8------|8-9----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-3----|1------|
S|12-----|8-9----|5------|
```

landings 2 and 4; no pick reset

095 4232

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-3----|2------|
S|8------|8-9----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-3----|2------|
S|12-----|8-9----|6------|
```

landings 2 and 4; no pick reset

096 4234

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-3----|4------|
S|8------|8-9----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-3----|4------|
S|12-----|8-9----|8------|
```

landings 2 and 4; no pick reset

Level 11 - Three-Position Relay

Set 9 of 9 - Exercises 097-108

1+2+1: note 1 in the first frame, notes 2-3 in the middle frame, note 4 in the third frame.

097 4241

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-4----|1------|
S|8------|8-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-4----|1------|
S|12-----|8-10---|5------|
```

landings 2 and 4; no pick reset

098 4242

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-4----|2------|
S|8------|8-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-4----|2------|
S|12-----|8-10---|6------|
```

landings 2 and 4; no pick reset

099 4243

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|2-4----|3------|
S|8------|8-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|2-4----|3------|
S|12-----|8-10---|7------|
```

landings 2 and 4; no pick reset

100 4312

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-1----|2------|
S|8------|9-7----|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-1----|2------|
S|12-----|9-7----|6------|
```

landings 2 and 4; no pick reset

101 4313

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-1----|3------|
S|8------|9-7----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-1----|3------|
S|12-----|9-7----|7------|
```

landings 2 and 4; no pick reset

102 4314

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-1----|4------|
S|8------|9-7----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-1----|4------|
S|12-----|9-7----|8------|
```

landings 2 and 4; no pick reset

103 4321

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-2----|1------|
S|8------|9-8----|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-2----|1------|
S|12-----|9-8----|5------|
```

landings 2 and 4; no pick reset

104 4323

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-2----|3------|
S|8------|9-8----|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-2----|3------|
S|12-----|9-8----|7------|
```

landings 2 and 4; no pick reset

105 4324

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-2----|4------|
S|8------|9-8----|12-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-2----|4------|
S|12-----|9-8----|8------|
```

landings 2 and 4; no pick reset

106 4341

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-4----|1------|
S|8------|9-10---|9------|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-4----|1------|
S|12-----|9-10---|5------|
```

landings 2 and 4; no pick reset

107 4342

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-4----|2------|
S|8------|9-10---|10-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-4----|2------|
S|12-----|9-10---|6------|
```

landings 2 and 4; no pick reset

108 4343

1+2+1 DOUBLE SHIFT

UP E>A>D>G>B>e P5>P7>P9

```
F|4------|3-4----|3------|
S|8------|9-10---|11-----|
```

DN e>B>G>D>A>E P9>P7>P5

```
F|4------|3-4----|3------|
S|12-----|9-10---|7------|
```

landings 2 and 4; no pick reset